Department of Computer Information Sciences and Mathematics

University of San Carlos

Cebu City, Philippines

IP TELEPHONY

A Networking Plan Documentation

Presented to Godwin Monserate

By

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Chapter I

Introduction

In today's society, networks have become a central medium of communication, enabling individuals to connect and communicate anytime and anywhere. This technology has revolutionized the transfer of information, making it more efficient and reliable. People can easily text and call each other at their convenience, fostering continuous interaction. Moreover, it has paved the way for establishing new business ventures and provided a platform for traditional businesses to expand their operations online. As businesses increasingly rely on digital connectivity to manage operations, collaborate across teams, and deliver services, robust network infrastructure has become essential.

Neon District Studio is a multimedia company that provides a wide range of multimedia services, including video production, animation, graphic design, and more. Its main branch is in Cebu City, with two additional branches in Manila and Davao. The company comprises several departments: Graphics Design Department, Animation Department, Video Production Department, Human Resource Department, Marketing Department, IT Department, Legal Department, and Management. Each department is equipped with local networks connected to Internet Service Providers (ISPs), facilitating both local communication and internet access. As the company grows and its operations expand, the demand for a stable, efficient, and secure network becomes more critical.

To address these needs, the group proposes establishing a robust and efficient network infrastructure tailored to the company's requirements. This plan aims to implement VLANs and inter-VLAN routing to enable seamless communication between departments, optimize network performance, and enhance security. Network redundancy will be introduced to mitigate traffic congestion and ensure continuity during failures. Additionally, Layer 2 security measures will be implemented to safeguard devices and data from unauthorized access. The plan also prioritizes the availability and

reliability of networks to maintain uninterrupted communication even during minor disruptions. Lastly, a control-based system for wireless network management will ensure efficient oversight and optimization of wireless resources. By adopting this network infrastructure, the company can enhance productivity, improve collaboration, and support its growth in a digitally connected world.

Chapter II.

Related Literature

The proposed network infrastructure plan aligns with the findings of Ledwaba, Pelser, and Fatoki (2019), who highlight the critical role of e-business applications in enhancing operational efficiency and delivering value propositions to customers. Their study emphasizes that businesses utilizing e-business applications, such as e-commerce and e-marketing, experience significant benefits in their operational processes, including improved communication and streamlined workflows. By implementing a robust and efficient network infrastructure, such as the one proposed in this plan, <company name> can create a solid foundation to support these applications effectively.

The proposed network's focus on reliability, security, and efficient communication channels directly facilitates the adoption of e-business solutions, enabling the company to deliver value propositions to customers efficiently. As noted in the study, e-business applications are relatively easy to implement and cost-effective for SMEs, making the establishment of a reliable network infrastructure a critical step toward achieving competitive advantages in today's digital-driven business landscape (Ledwaba et al., 2019).

The study by Malasowe et al. (2024) underscores the transformative potential of VoIP-based telephony systems in enhancing network scalability and communication efficiency. Their research highlights how IP-telephony technologies integrate voice, video, and data solutions on a unified network, addressing critical challenges such as packet loss, latency, and jitter. These findings are highly relevant to the proposed network infrastructure plan, as they emphasize the importance of robust network design for achieving efficient data exchange and reliable communication.

In the context of Neon District Studio, implementing a well-structured network infrastructure with VLANs, redundancy, and Layer 2 security, as proposed, can support advanced technologies like VoIP to enhance internal communication. Adopting Ethernet-cabled networks, wireless LANs, and cloud

storage—recommended by Malasowe et al.—further strengthens the system's capability to provide seamless communication, mobility, and productivity. This integration is particularly beneficial for ensuring uninterrupted access to critical resources and fostering collaboration across the company's departments, aligning to create a scalable and efficient network.

The importance of collaborative practices in video editing has been emphasized in recent research, particularly as the industry evolves to incorporate remote work and digital-first workflows. Okopnyi (2024) highlights that although traditionally a solo activity, video editing increasingly involves multiple collaborators in professional settings. His study identifies key strategies such as organizational mechanisms, documentation, and iconic referencing, essential for managing uncertainties and facilitating stakeholder agreements. These findings underscore the need for effective tools and environments to support these practices.

Additionally, the shift to remote work during the COVID-19 pandemic has had significant implications for video production. Okopnyi's research notes that while remote work introduces challenges like communication barriers, it also offers opportunities for enhanced accessibility and collaboration. These insights align with trends in networked work environments, where reliable connectivity and resource sharing become critical for productivity.

To address these needs, Okopnyi proposes three design approaches for collaborative video editing tools:Holistic Design,Tailored Solutions,Configurable Features.

These principles can be directly related to the proposed network infrastructure for Neon District Studio. By implementing technologies such as VLANs for segmentation, redundancy for reliability, and secure wireless access, the network can create a foundation that supports the collaborative demands of multimedia production. Furthermore, enabling remote access to resources and incorporating cloud-based solutions aligns with the evolving needs of video editors to work collaboratively in dispersed teams.

In conclusion, the research highlights that robust and flexible network systems are indispensable for fostering video editing and production collaboration. Integrating these insights into the proposed infrastructure plan will ensure that Neon District Studio is equipped to support modern, collaborative workflows effectively.

Chapter III.

Technical Background

This chapter encompasses the various tools utilized by the group in developing the proposed network infrastructure. These tools were chosen to ensure accurate planning, testing, and documentation for the successful implementation of the network.

Network Simulation Tool

A network simulation tool allows for the simulation of the proposed network infrastructure to test for potential issues or faults before physical implementation. This ensures a seamless and successful deployment of the network in the company.

Example:

- **Cisco Packet Tracer**: A network simulation tool developed by Cisco, containing network devices designed by the company. The group selected this tool due to its compatibility with the Cisco devices proposed for the network and the team's high familiarity with its features.

Floor Plan Creation Tool

A floor plan creation tool enables visualization of the company's building layout. It helps in planning the placement of networking devices, cables, and end-user devices, ensuring optimal positioning for efficient network functionality.

Example:

- **Lucidchart:** A floor plan creation tool equipped with a library of image icons for devices and user-friendly controls for creating building layouts. This tool was used to visualize and plan the company's network infrastructure.

Spreadsheet Tool

A spreadsheet tool assists in organizing and visualizing data. It also provides basic computational functionalities, which are critical for planning and managing network-related data.

Example:

- **Google Sheets:** A cloud-based spreadsheet tool by Google used for computing and organizing the addressing scheme for the network plan. Its collaborative features allow multiple users to work simultaneously, improving efficiency and data accuracy.

Word Processing Tool

A word processing tool is essential for documenting the details of the network infrastructure proposal, including plans, configurations, and operational guidelines.

Example:

- Google Docs: A cloud-based word processing tool by Google. It was used to draft and edit the documentation of the proposed network plan. The collaboration features of Google Docs made it easier for the group to share and review content in real-time.

Chapter IV

Methodology

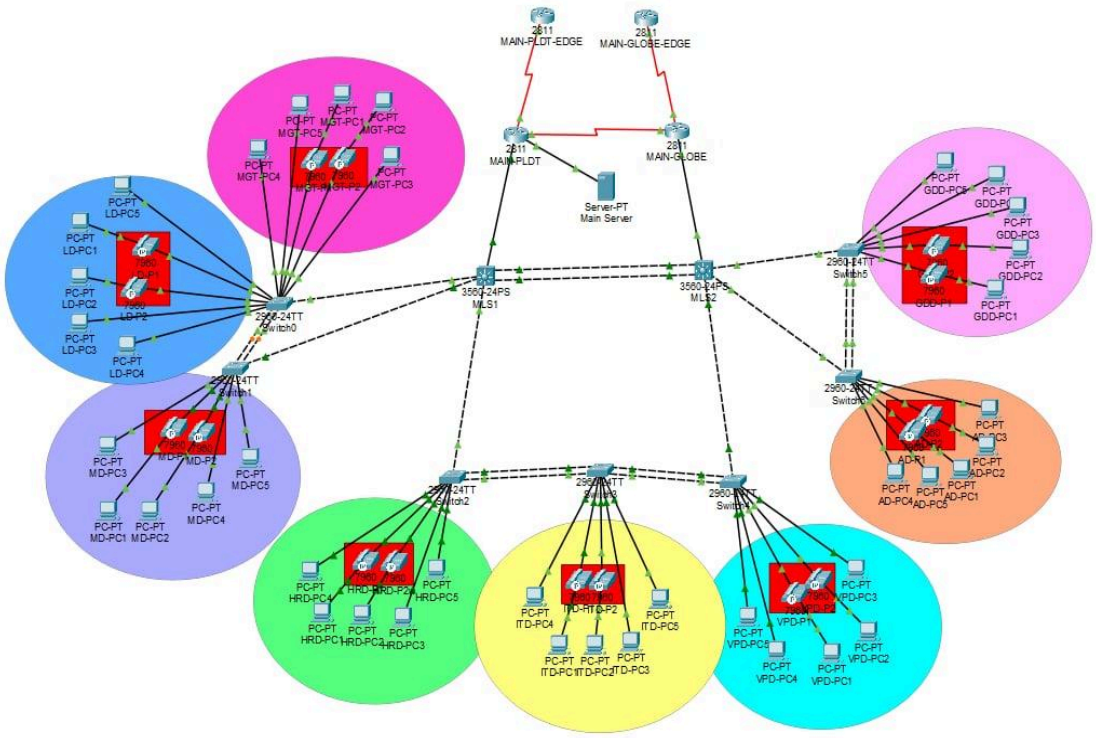
This chapter covers the floor plans proposed for each building, the addressing schemes, and the protocols that will be implemented.

Main Building

The main building in Cebu will be a 3-floor building with 8 departments: graphics design, animation, video production, human resources, marketing, IT, legal, and management. The building will have 2 routers, 2 multilayer switches, and 7 switches. A dynamic routing protocol will be implemented to allow connections to other branches. Vlans and InterVlan routing will be used to separate and allow communication between different departments. Redundancy protocols will also be implemented to improve the network performance and a layer 2 security for network security.

Logical Topology

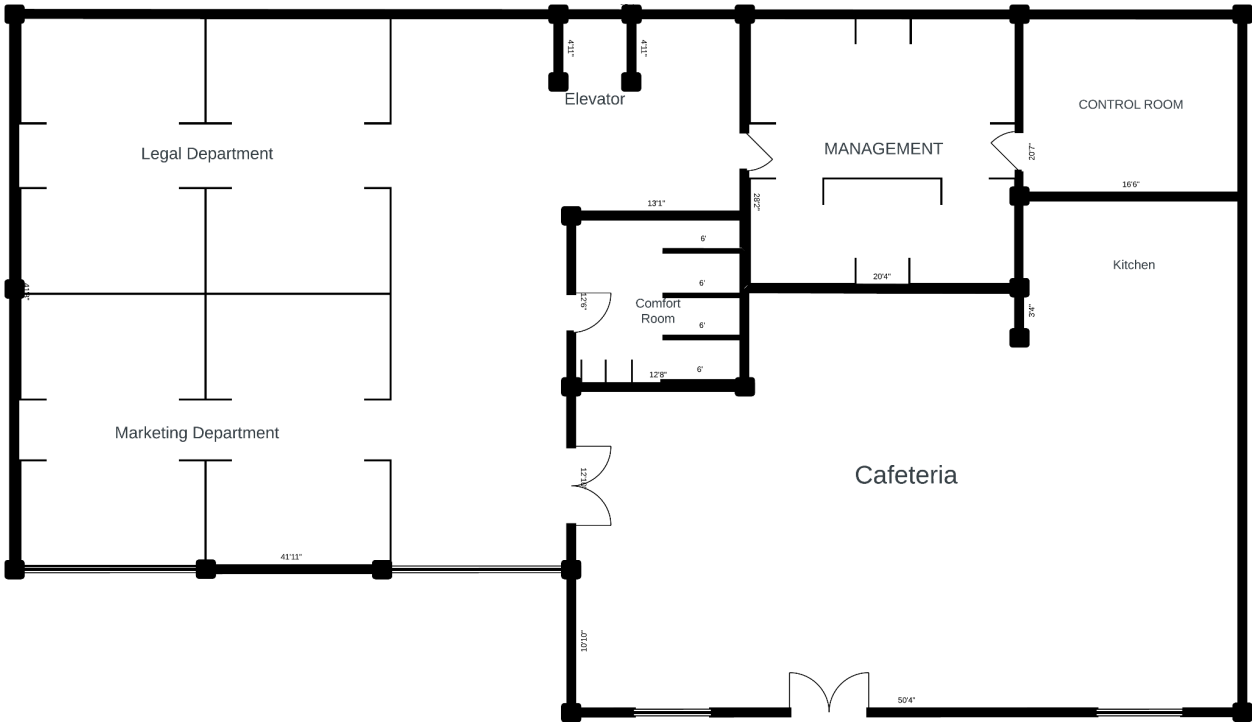
Figure 1.0: This highlights the logical topology that was designed in the Packet Tracer simulation. The logical topology provides a comprehensive representation of the various departments within the network and their associated IP phones, each mapped to their respective VLANs for efficient communication and management.

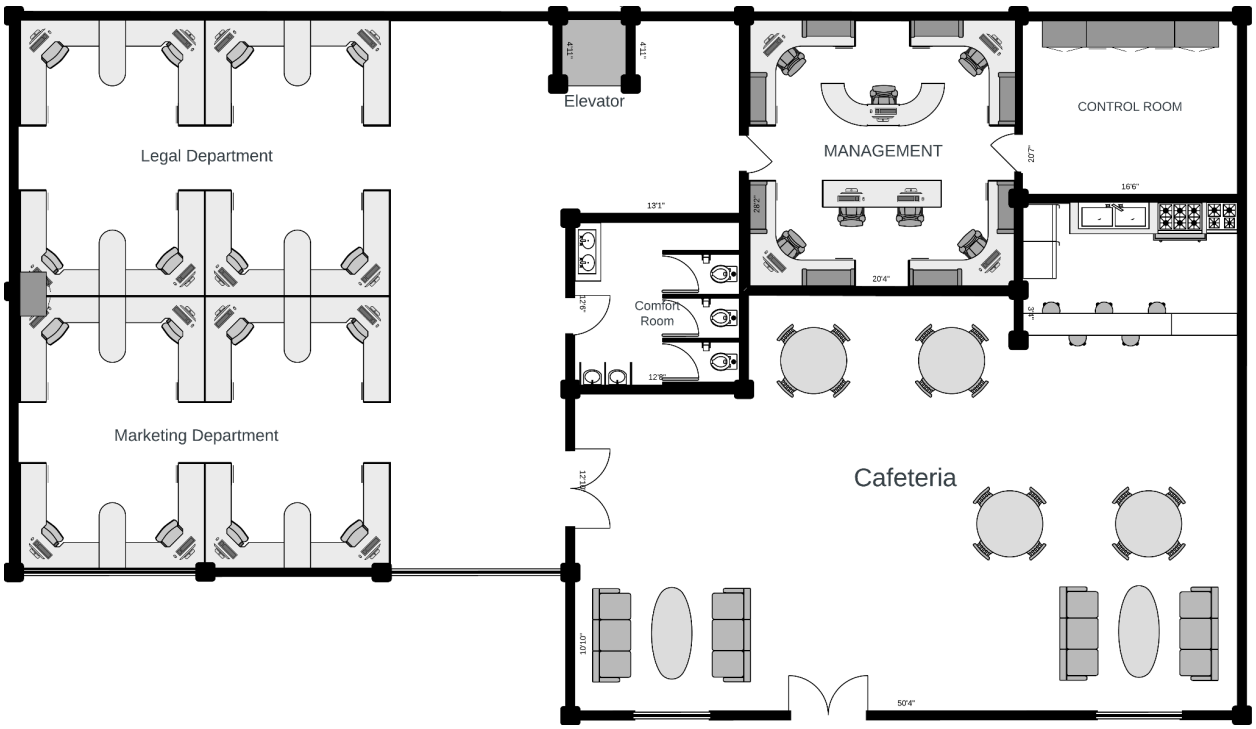


Physical Topology

1st Floor

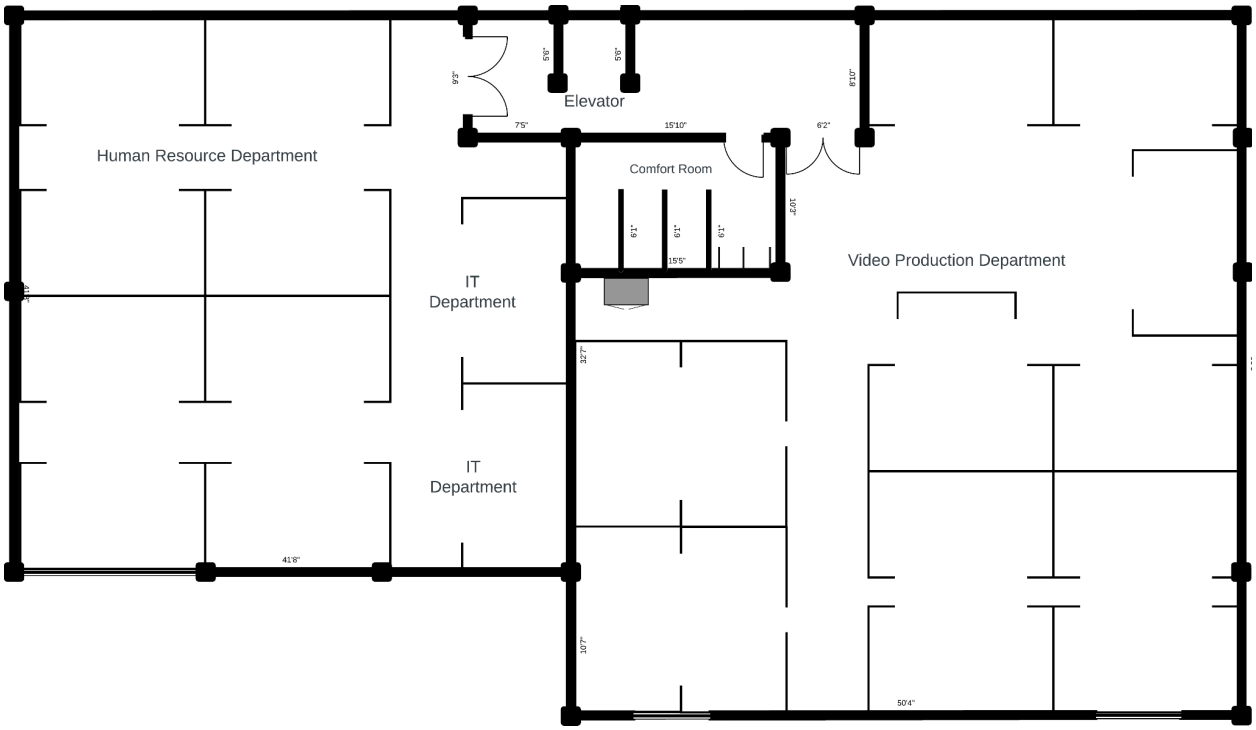
Figure 1.1 - 1.2: The first floor of the main building comprises three departments: Legal, Marketing, and Management. It also features amenities for workers, including a cafeteria, kitchen, and restroom. An elevator is strategically placed to facilitate movement between floors. This floor also houses the server and all the building's routers. Additionally, a generic container at the center holds the switches that connect the departments to the server room. The different department's room and cubicles are properly proportioned to 1:1 ratio giving an equal and ample space to work comfortably.

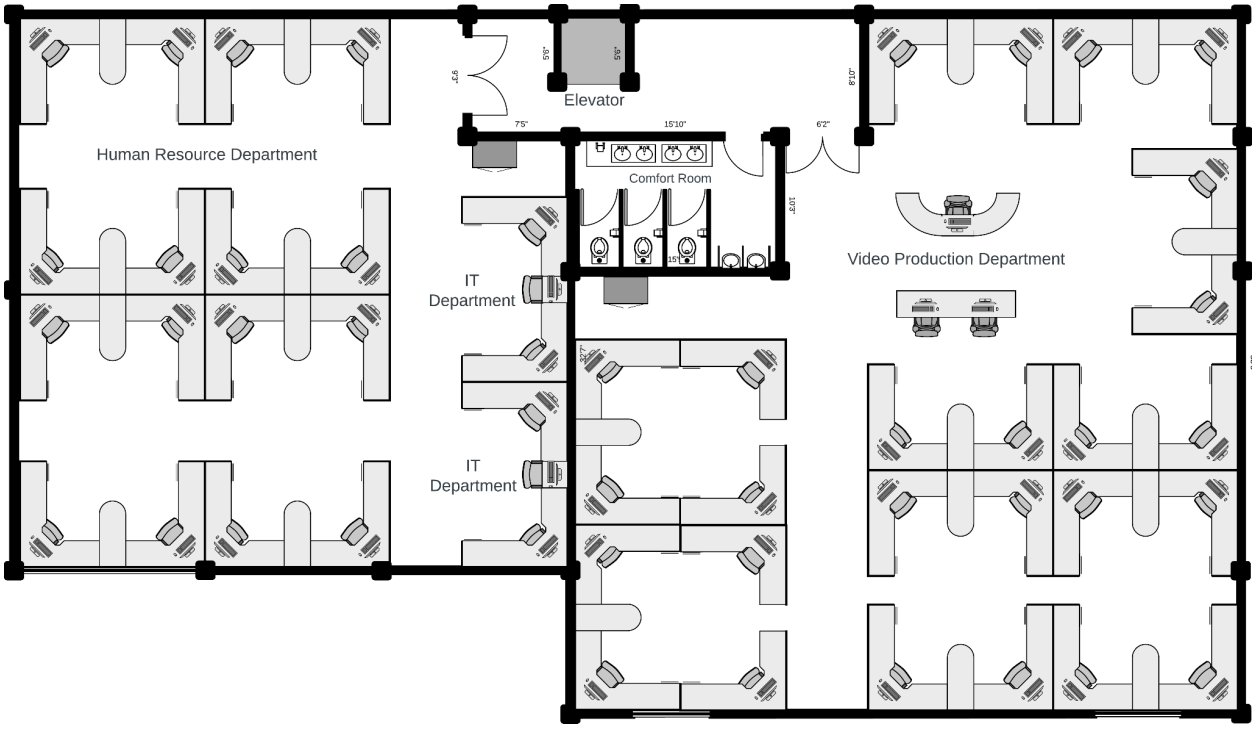

Legal Department
Marketing Department
Elevator
4'11"
4'11"
MANAGEMENT
CONTROL ROOM
20'7"
16'6"
13'1"
28'2"
6'
6'
6'
6'
12'8"
20'4"
Comfort Room
12'6"
Kitchen
3'4"
Cafeteria
12'7"
41'11"
10'10"
50'4"


Legal Department
Marketing Department
Elevator
4'11"
4'11"
MANAGEMENT
CONTROL ROOM
20'7"
16'6"
13'1"
28'2"
20'4"
Comfort Room
12'6"
12'8"
Cafeteria
12'7"
10'10"
50'4"

2nd Floor

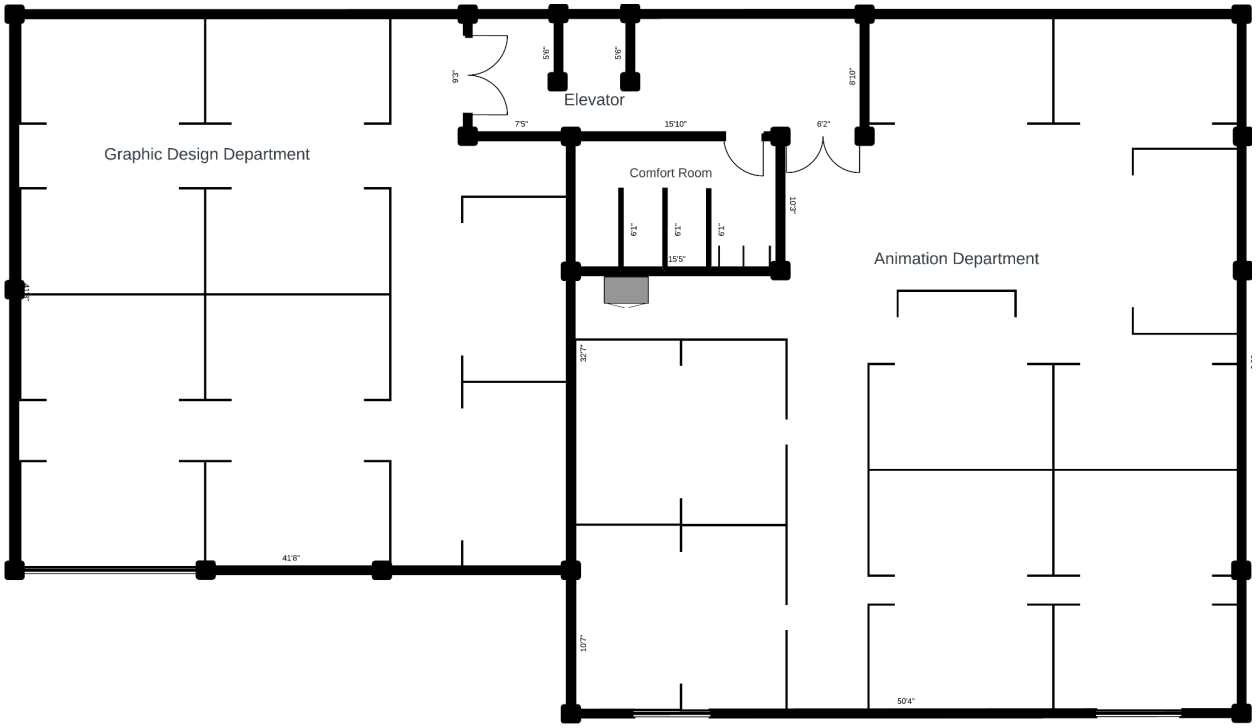
Figure 1.3-1.4: The second floor of the main building includes three departments: Human Resources, IT, and Video Production. Additionally, this floor houses a central container with switches connecting all three departments to the server room. The HR Department consists of two large cubicles, each containing four small cubicles. Each large cubicle maintains a 1:1 ratio with its small cubicles, but the combined structure forms a 1:2 ratio. The IT Department is deliberately designed with a disproportionate layout to ensure strategic placement. Located on the second floor, this positioning allows for efficient mobility in case of device issues and facilitates the maintenance of devices across all departments. The Video Production Department occupies the largest area on the second floor, reflecting its importance as one of the company's core services. Its layout features disproportionately arranged cubicles, strategically placed to optimize the department's functionality and workspace utilization.

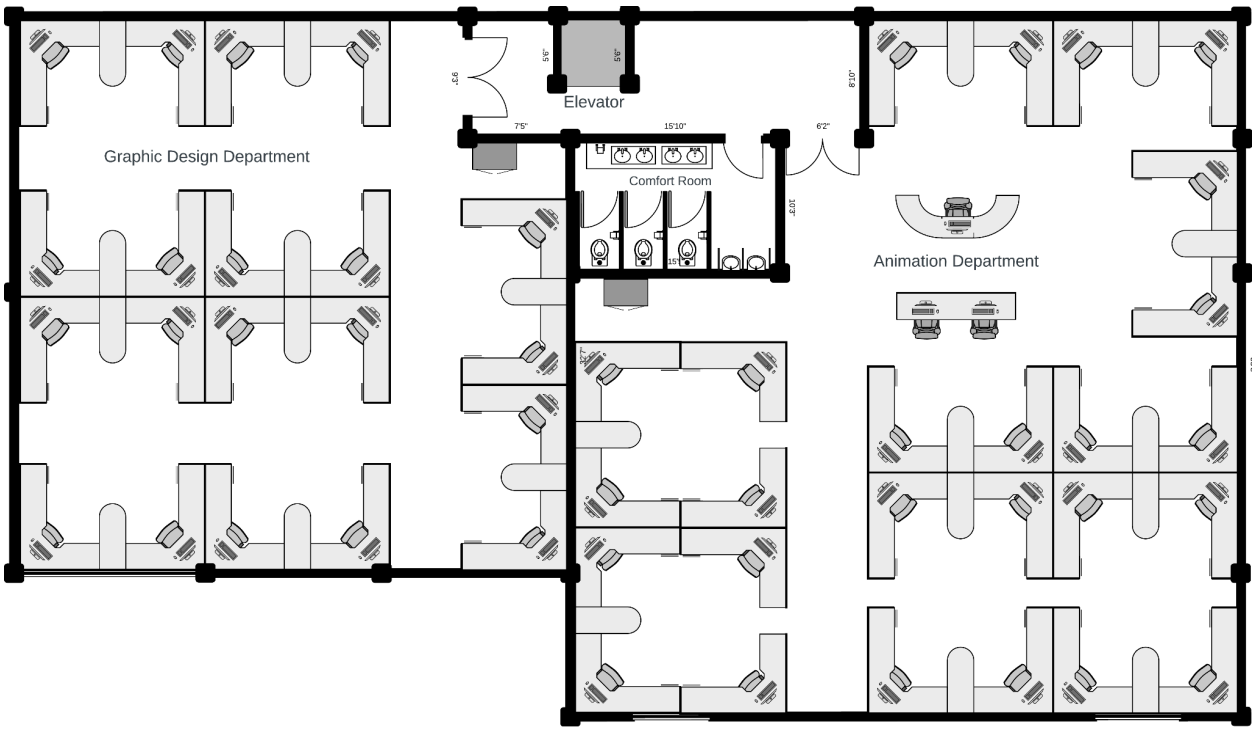




3rd Floor

Figure 1.5 - 1.6: The Third floor of the main building includes two departments: Graphic Design Department and Animation Department. Additionally, this floor houses the generic container in the center containing the switches that connect the three departments to the server room.

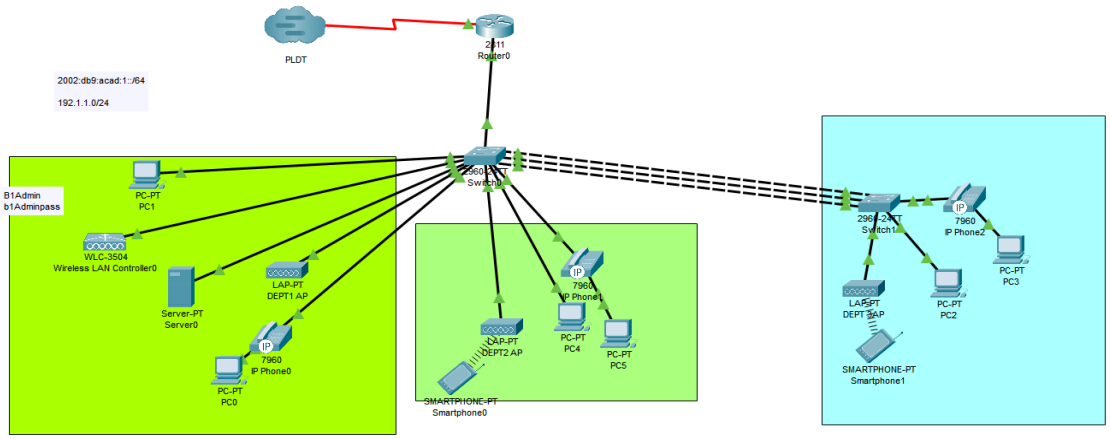

Graphic Design Department
Elevator
Comfort Room
Animation Department
9'6"
5'6"
5'6"
8'10"
7'5"
15'10"
6'2"
10'7"
6'1"
6'1"
6'1"
15'5"
32'7"
41'8"
10'7"
50'4"


Graphic Design Department
Elevator
Comfort Room
Animation Department
9'7"
5'6"
5'6"
8'10"
7'5"
15'10"
6'2"
10'3"
15'5"

Branch 1 - Manila

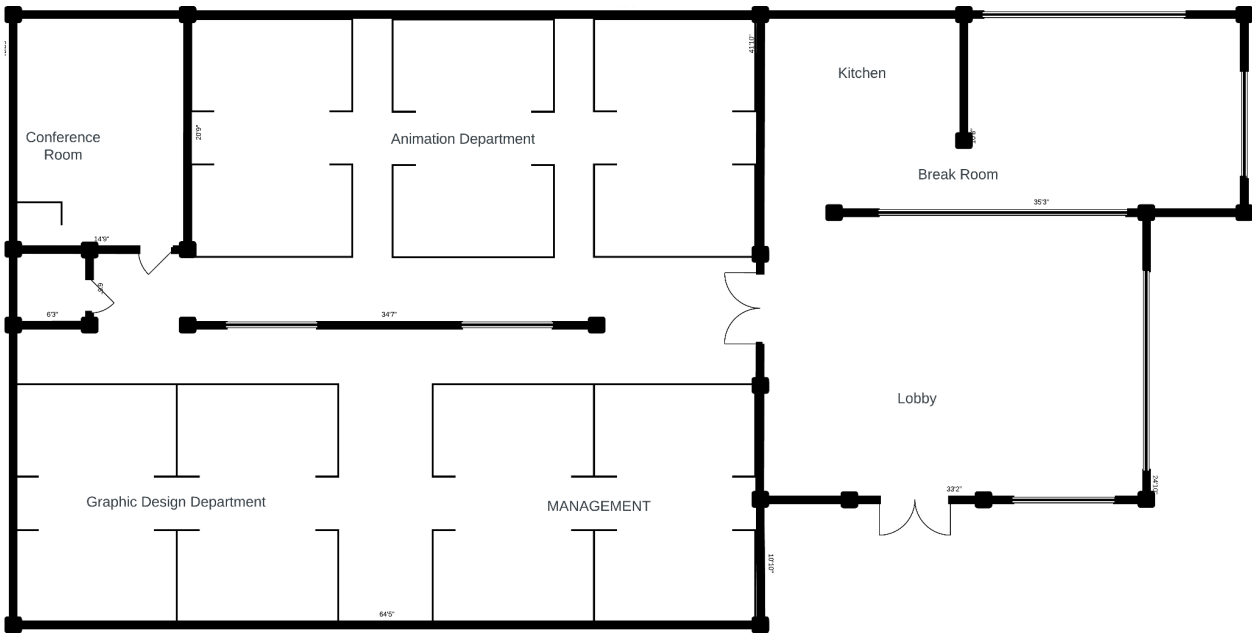
Logical Topology

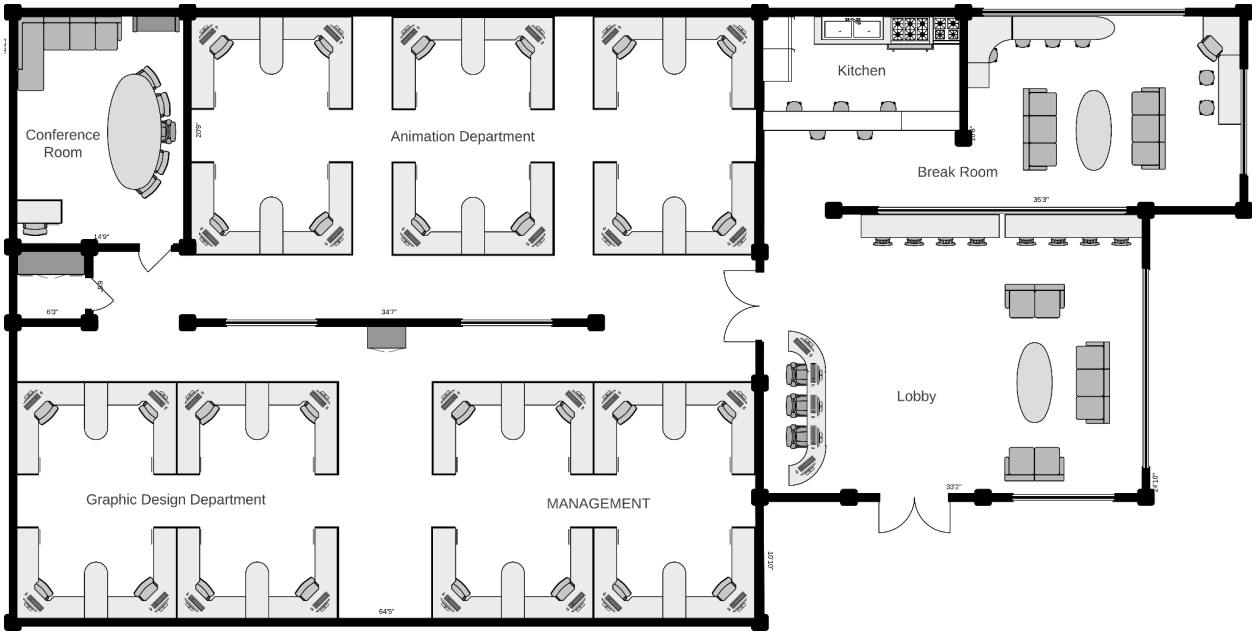
Figure 2.0: Branch 1 in Manila features a single-floor building with three departments: Management, Animation, and Graphics Design. Each department is equipped with one IP phone and at least two PCs. This includes one router, which serves as the DHCP server for local networks, and two switches for departmental connectivity. A dedicated server is configured as a AAA TACACS server and a web server. The network includes a controlled wireless environment to support mobility and wireless devices. VLANs and Inter-VLAN routing are implemented to separate departmental networks while enabling communication. Additionally, a dynamic routing protocol connects the branch to other locations, and redundancy protocols are implemented to enhance the network



Physical Topology

Figure 2.1-2.2: Branch 1 in Manila is a single-floor building housing three departments: Management, Animation, and Graphics Design. The floor plan also includes a conference room for meetings and amenities such as a kitchen, break room, and lobby to support employee needs and comfort.

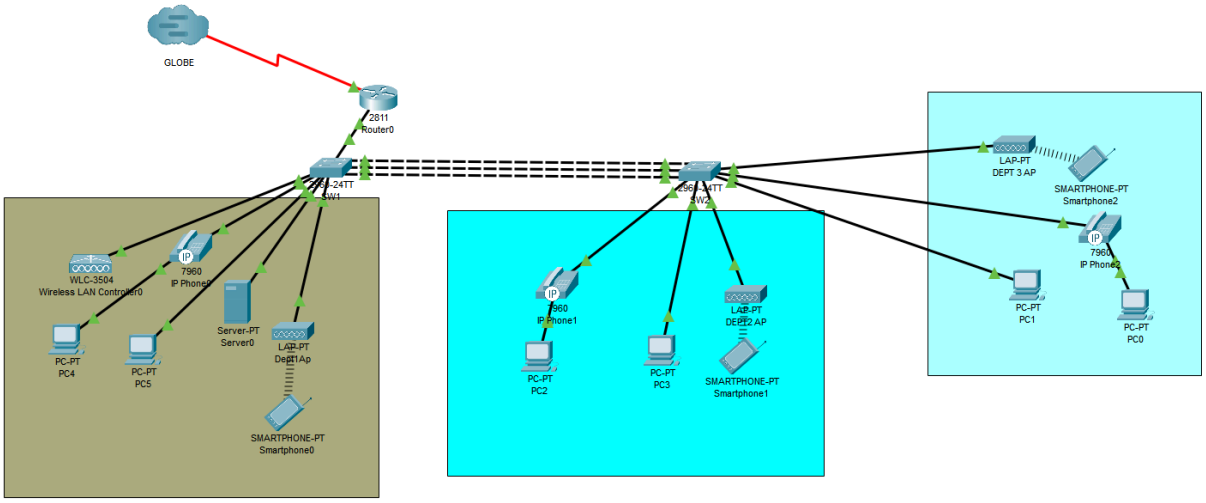




Branch 2 - Davao

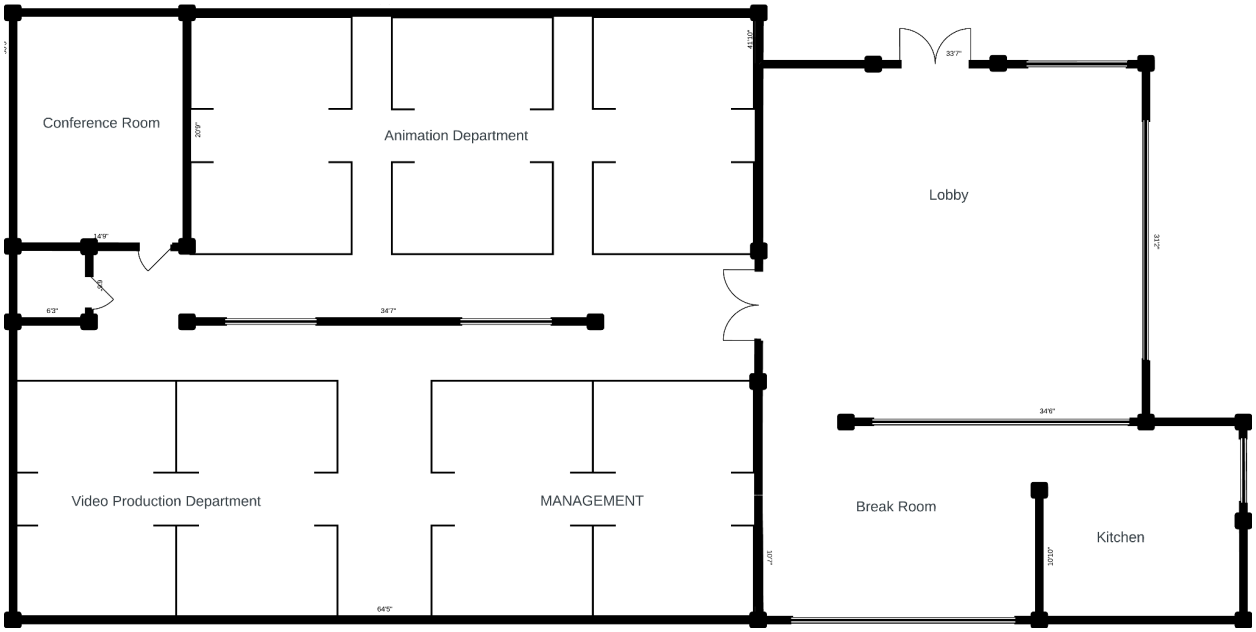
Logical Topology

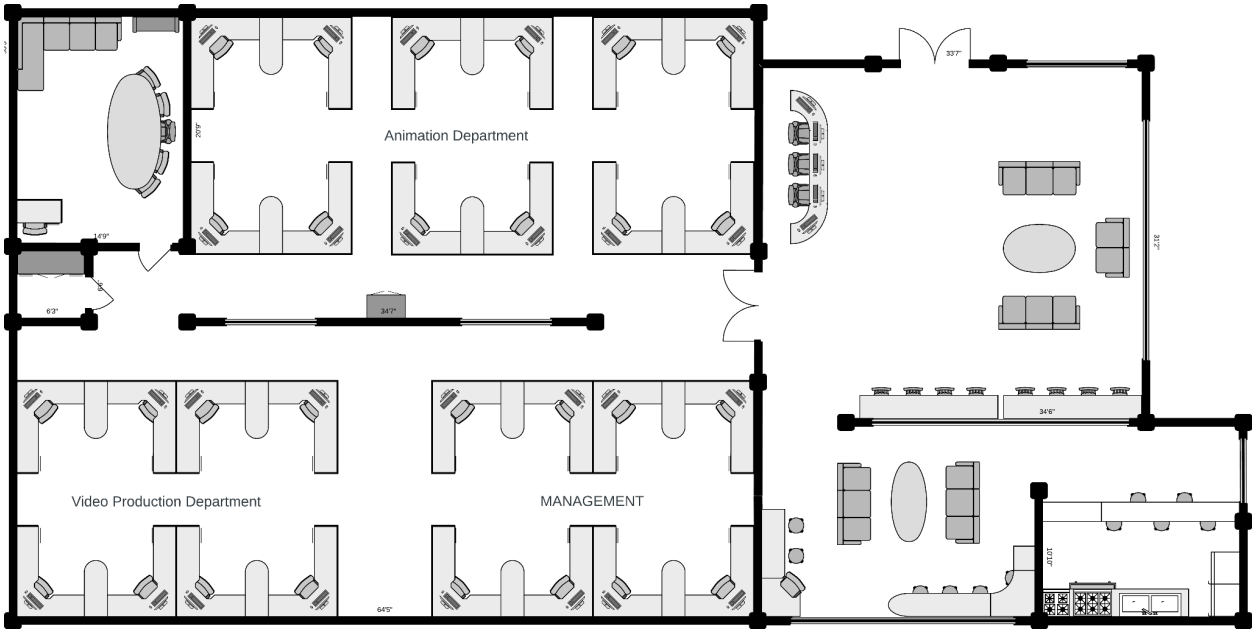
Figure 3.0: This figure shows the floor plan proposed for Branch 2 in Davao. Same as the Manila branch, A 1-floor building was created consisting of three departments. Each will contain A IP phone and at least two PCs. It will also have a single router acting as DHCP and two switches. A controlled wireless network environment will also be present in this building and A server that acts as a AAA tacacs server and a web server. A dynamic .



Logical Topology

Figure 3.1-3.2: Branch 2 in Davao is a single-story building consisting of three departments: Management, Animation, and Graphics Design. The floor plan also includes a conference room for meetings, along with amenities such as a kitchen, break room, and lobby to ensure employee comfort and convenience.


Conference Room
Animation Department
Lobby
Video Production Department
MANAGEMENT
Break Room
Kitchen


Animation Department
Video Production Department
MANAGEMENT

IP Addressing

The group makes use of the CIDR addressing scheme to calculate the addresses and assign them to different networks.

BUILDING	Networks	ipv4 address(classless)	ipv6 address
MAIN BRANCH	Legal	172.168.1.0/24	172:ACAD:DB8:10::/64
	Animation	172.168.2.0/24	172:ACAD:DB8:20::/64
	VideoProduction	172.168.3.0/24	172:ACAD:DB8:30::/64
	HR	172.168.4.0/24	172:ACAD:DB8:40::/64
	Marketing	172.168.5.0/24	172:ACAD:DB8:50::/64
	IT	172.168.6.0/24	172:ACAD:DB8:60::/64
	Legal	172.168.7.0/24	172:ACAD:DB8:70::/64
	Management	172.168.8.0/24	172:ACAD:DB8:99::/64
	Voice	172.168.9.0/24	172:ACAD:DB8:150::/64
	Server	172.168.10.0/24	172:ACAD:DB8:100::/64
	MAIN-PLDT -> MAIN-PLDT-EDGE	172.168.11.0/24	172:ACAD:DB8:110::/64
	MAIN-GLOBE-> MAIN-GLOBE-EDGE	172.168.12.0/24	172:ACAD:DB8:120::/64
	MAIN-PLDT ->MAIN-GLOBE	172.168.13.0/24	172:ACAD:DB8:130::/64
	MAIN-PLDT-EDGE -> ISP	10.10.10.0/30	2001:DB8:ACAD:FFF::/64
	MAIN-GLOBE-EDGE -> ISP	11.11.11.0/30	2001:DB8:ACAD:FFFF::/64
Branch1 - Manila			

	Management	192.1.1.0/24	2002:DB9:ACAD:1::/64
	Animation	192.1.2.0/24	2002:DB9:ACAD:2::/64
	Graphic Design	192.1.3.0/24	2002:DB9:ACAD:3::/64
	Voice	192.1.4.0/24	2002:DB9:ACAD:4::/64
	Branch1 to PLDT	60.60.60.0/30	2001:DB8:ACAD:FF::/64
Branch2 - Davao			
	Management	192.2.2.0/24	2001:DB8:ACAD:2::/64
	Animation	192.2.3.0/24	2001:DB8:ACAD:3::/64
	Video Production	192.2.4.0/24	2001:DB8:ACAD:4::/64
	Voice	192.2.5.0/24	2001:DB8:ACAD:5::/64
	Wireless/admin	192.2.1.0/24	2001:DB8:ACAD:1::/64
	Branch2 to GLOBE	50.50.50.0/30	2001:DB8:ACAD:F::/64
GLOBE			
	GLOBE network	101.101.101.0 /30	2008:DB8:ACAD:BFF::/64
PLDT			
	PLDT network	100.100.100.0/30	1001:DBA:ACAD:AFF::/64

Chapter V.

Discussion and Implementation

Main Branch

Nine Vlans were created in this building, namely Vlan 10 (Legal Department), Vlan 20 (Marketing Department) Vlan 30 (Human Resource Department) Vlan 40 (IT Department) Vlan 50 (Video Production Department) Vlan 60 (Graphic Design Department) Vlan 70 (Animations Department) Vlan 99 (Management) Vlan 150(Voice).Each department was able to communicate with each other using router-on-a-stick implementation for Inter-Vlan routing. LACP was implemented as etherchannel for the switches for redundancy purposes and the router was implemented with DCHPv4 and DHCPv6 with SLAAC to dynamically assign addresses to end devices. OSPF was implemented as the routing protocol and going out to the local network EIGRP redistribution was implemented since the ISP uses EIGRP protocol to connect to other branch. Port-Securities with violation restrict and unused ports shut down were implemented on all switches. SSH is also implemented in switches routers with AAA authentication with tacacs configured. The branch server was able to host HTTP, FTP, and Email services.

Branch 1 - Manila

Four Vlans were created in this building, specifically Vlan 10(Management), Vlan 20(Animation), Vlan 30(GraphicsDesign), Vlan 150 (<Voice>). Each department was able to communicate with each other using router-on-a-stick implementation for Inter-Vlan routing. LACP was implemented as etherchannel for the switches for redundancy purposes and the router was implemented with DCHPv4 and DHCPv6 with SLAAC to dynamically assign addresses to end devices. EIGRP was implemented as the routing protocol since the ISP(PLDT) connected to this branch uses the same routing protocol. Port-Securities with violation restrict and unused ports shut down were implemented on all

switches. SSH is also implemented in switches routers with AAA authentication with tacacs configured. The branch server was able to host HTTP, FTP, and Email services.

Branch 2 - Davao

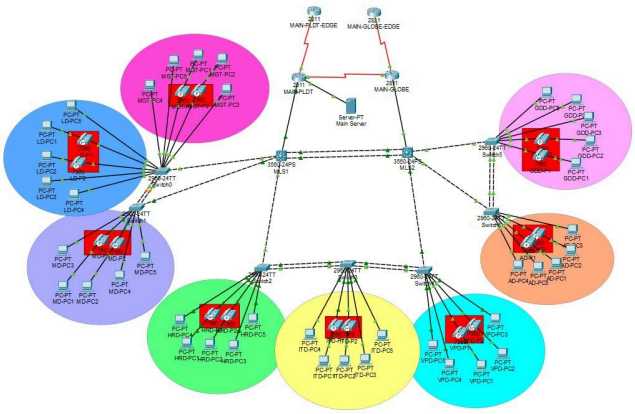
Five Vlans were implemented in this building namely Vlan 1(Wireless/admin), Vlan 10(Management), Vlan20(GraphicsDesign), Vlan 30(Marketing), Vlan 150(voice). The departments were able to communicate using router-on-a-stick inter-vlan routing. LACP was implemented as an etherchannel for the switches. The router was also implemented with DHCPv4 and DHCPv6 with SLAAC to dynamically assign IP addresses to end devices. EIGRP was also used as a Routing Protocol since the ISP(GLOBE) uses EIGRP enabling communication to other branches. Switches and Routers are configured with SSH and AAA authentication with tacacs server. Port security with violation restrict and all unused ports were shut down for all switches. The branch server was also able to host HTTP,FTP,and Email services.

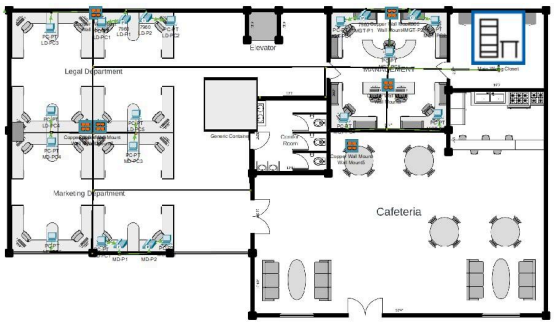
Chapter VI.

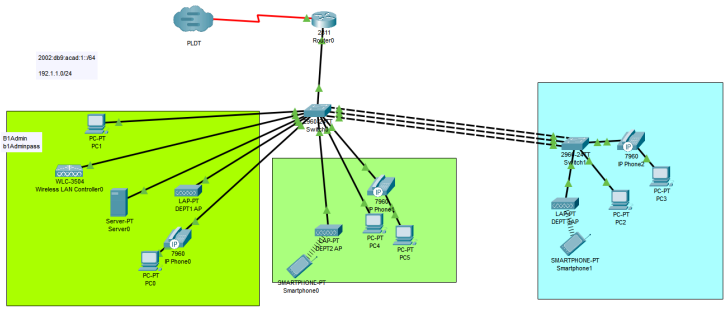
Conclusion and Recommendation

In conclusion, the proposed network architecture can be effectively implemented within Neon District Studios. The network will enhance their operations by improving communication and collaboration across various branches. Since the company relies heavily on technology for content creation, distribution, and client engagement, a robust network infrastructure is essential. Additionally, leveraging modern computing technologies, such as cloud services, can provide Neon District Studios with scalable, flexible, and cost-efficient solutions. This network will not only support the studio's current operations but also ensure it is prepared for future growth and technological advancements, ultimately driving increased productivity and business success.

Appendices




Legal Department
Elevator
Marketing Department
Cafeteria


PLDT
Router0
2002:db9:acad:1::/64
192.1.1.0/24
B1Admin
b1Adminpass
PC-PT
PC1
WLC-3504
Wireless LAN Controller0
Server-PT
Server0
LAP-PT
DEPT1AP
7960
IP Phone0
PC-PT
PC9
LAP-PT
DEPT2AP
7960
IP Phone1
PC-PT
PC4
PC-PT
PC5
SMARTPHONE-PT
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SMARTPHONE-PT
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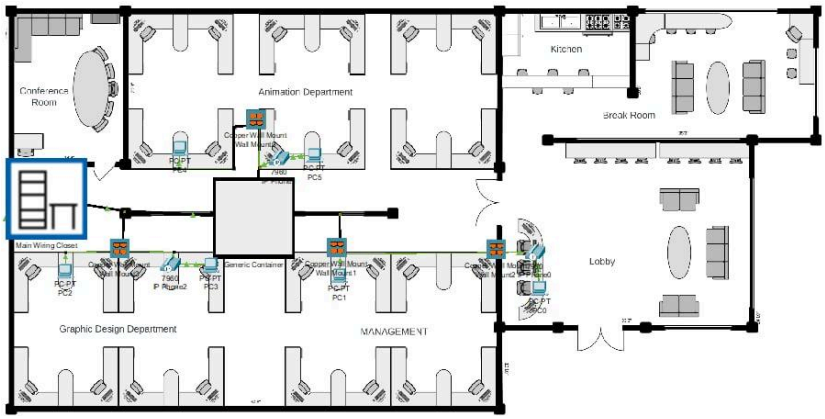
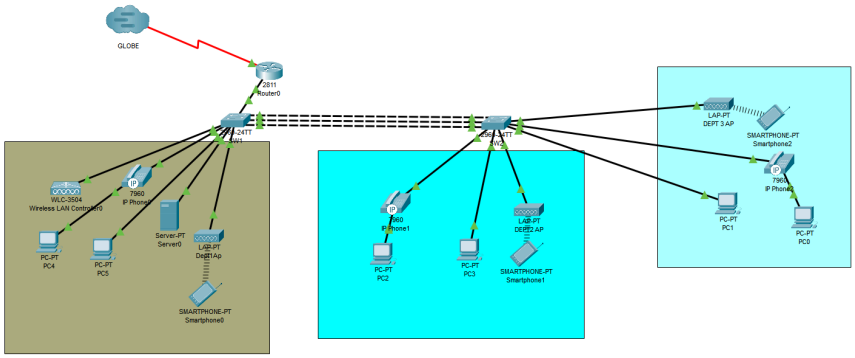

Conference Room
Animation Department
Kitchen
Break Room
Main Wiring Closet
Generic Container
Lobby
Graphic Design Department
MANAGEMENT


GLOBE
2811
Router0
2960-24TT
Switch1
WLC-3504
Wireless LAN Controller0
7960
IP Phone0
Server-PT
Server0
LAP-PT
Dept1Ap
PC-PT
PC4
PC-PT
PC5
SMARTPHONE-PT
Smartphone0
2960-24TT
Switch0
7960
IP Phone1
PC-PT
PC2
PC-PT
PC3
LAP-PT
DEPT 2 AP
SMARTPHONE-PT
Smartphone1
LAP-PT
DEPT 3 AP
SMARTPHONE-PT
Smartphone2
7960
IP Phone2
PC-PT
PC1
PC-PT
PC0

Codes

Main Branch

MAIN-PLDT && MAIN GLOBE ROUTER CONFIGURATION

```
telephony-service
max-ephones 16
max-dn 16
ip source-address 172.168.9.2 port 2000
auto assign 1 to 16
exit

ephone-dn 1
number 17201
exit

ephone-dn 2
number 17202
exit

ephone-dn 3
number 17203
exit

ephone-dn 4
number 17204
exit

ephone-dn 5
number 17205
exit

ephone-dn 6
number 17206
```

```
exit

ephone-dn 7
number 17207
exit

ephone-dn 8
number 17208
exit

ephone-dn 9
number 17209
exit

ephone-dn 10
number 17210
exit

ephone-dn 11
number 17211
exit

ephone-dn 12
number 17212
exit

ephone-dn 13
number 17213
exit

ephone-dn 14
number 17214
exit

ephone-dn 15
```

```
number 17215
exit

ephone-dn 16
number 17216
exit

ip dhcp pool PHONE-DHCP
network 172.168.9.0 255.255.255.0
default-router 172.168.9.1
option 150 ip 172.168.9.1
exit

ipv6 dhcp pool LD
address prefix 172:ACAD:DB8:10::0/64
dns-server 172:ACAD:DB8:100::2
domain-name www.neondistrictstudio.com
exit

ipv6 dhcp pool MD
address prefix 172:ACAD:DB8:20::0/64
dns-server 172:ACAD:DB8:100::2
domain-name www.neondistrictstudio.com
exit

ipv6 dhcp pool HRD
address prefix 172:ACAD:DB8:30::0/64
dns-server 172:ACAD:DB8:100::2
domain-name www.neondistrictstudio.com
exit

ipv6 dhcp pool ITD
address prefix 172:ACAD:DB8:40::0/64
dns-server 172:ACAD:DB8:100::2
domain-name www.neondistrictstudio.com
```

```
exit

ipv6 dhcp pool VPD
address prefix 172:ACAD:DB8:50::0/64
dns-server 172:ACAD:DB8:100::2
domain-name www.neondistrictstudio.com
exit

ipv6 dhcp pool GDD
address prefix 172:ACAD:DB8:60::0/64
dns-server 172:ACAD:DB8:100::2
domain-name www.neondistrictstudio.com
exit

ipv6 dhcp pool AD
address prefix 172:ACAD:DB8:70::0/64
dns-server 172:ACAD:DB8:100::2
domain-name www.neondistrictstudio.com
exit

ipv6 dhcp pool MGT
address prefix 172:ACAD:DB8:80::0/64
dns-server 172:ACAD:DB8:100::2
domain-name www.neondistrictstudio.com
exit

ipv6 dhcp pool PHONE-DHCPv6
address prefix 172:ACAD:DB8:99::0/64
dns-server 172:ACAD:DB8:100::2
domain-name www.neondistrictstudio.com
exit

router ospf 10
network 172.168.1.0 0.0.0.255 area 1
network 172.168.2.0 0.0.0.255 area 1
```

```
network 172.168.3.0 0.0.0.255 area 1
network 172.168.4.0 0.0.0.255 area 1
network 172.168.5.0 0.0.0.255 area 1
network 172.168.6.0 0.0.0.255 area 1
network 172.168.7.0 0.0.0.255 area 1
network 172.168.8.0 0.0.0.255 area 1
network 172.168.9.0 0.0.0.255 area 1
network 172.168.10.0 0.0.0.255 area 1
network 172.168.13.0 0.0.0.255 area 1
network 172.168.11.0 0.0.0.255 area 0
area 1 range 172.168.1.0 255.255.240.0

int se0/0/0
ip address 172.168.11.0 255.255.255.0
no shutdown
exit

int fa0/1
ip address 172.168.10.1 255.255.255.0
no shutdown
exit

int fa0/0
no shutdown
exit

int fa0/0.10
encapsulation dot1Q 10
ip address 172.168.1.2 255.255.255.0
ip helper-address 172.168.10.2 255.255.255.0
standby 1 ip 172.168.1.1
standby 1 priority 105
standby 1 preempt
standby 1 track Serial0/0/0
ipv6 enable
```

```
ipv6 add fe80::1 link-local
ipv6 add 172:DB8:ACAD:10::2/64
ipv6 nd other-config-flag
ipv6 dhcp server LD
exit

int fa0/0.20
encapsulation dot1Q 20
ip address 172.168.2.2 255.255.255.0
ip helper-address 172.168.10.2 255.255.255.0
standby 1 ip 172.168.2.1
standby 1 priority 105
standby 1 preempt
standby 1 track Serial0/0/0
ipv6 enable
ipv6 add fe80::1 link-local
ipv6 add 172:DB8:ACAD:20::2/64
ipv6 nd other-config-flag
ipv6 dhcp server MD
exit

int fa0/0.30
encapsulation dot1Q 30
ip address 172.168.3.2 255.255.255.0
ip helper-address 172.168.10.2 255.255.255.0
standby 1 ip 172.168.3.1
standby 1 priority 105
standby 1 preempt
standby 1 track Serial0/0/0
ipv6 enable
ipv6 add fe80::1 link-local
ipv6 add 172:DB8:ACAD:30::2/64
ipv6 nd other-config-flag
ipv6 dhcp server HRD
exit
```

```
int fa0/0.40
encapsulation dot1Q 40
ip address 172.168.4.2 255.255.255.0
ip helper-address 172.168.10.2 255.255.255.0
standby 1 ip 172.168.4.1
standby 1 priority 105
standby 1 preempt
standby 1 track Serial0/0/0
ipv6 enable
ipv6 add fe80::1 link-local
ipv6 add 172:DB8:ACAD:40::2/64
ipv6 nd other-config-flag
ipv6 dhcp server ITD
exit

int fa0/0.50
encapsulation dot1Q 50
ip address 172.168.5.2 255.255.255.0
ip helper-address 172.168.10.2 255.255.255.0
standby 1 ip 172.168.5.1
standby 1 priority 105
standby 1 preempt
standby 1 track Serial0/0/0
ipv6 enable
ipv6 add fe80::1 link-local
ipv6 add 172:DB8:ACAD:50::2/64
ipv6 nd other-config-flag
ipv6 dhcp server VPD
exit

int fa0/0.60
encapsulation dot1Q 60
```

```
ip address 172.168.6.2 255.255.255.0
ip helper-address 172.168.10.2 255.255.255.0
standby 1 ip 172.168.6.1
standby 1 priority 105
standby 1 preempt
standby 1 track Serial0/0/0
ipv6 enable
ipv6 add fe80::1 link-local
ipv6 add 172:DB8:ACAD:60::2/64
ipv6 nd other-config-flag
ipv6 dhcp server GDD
exit


int fa0/0.70
encapsulation dot1Q 70
ip address 172.168.7.2 255.255.255.0
ip helper-address 172.168.10.2 255.255.255.0
standby 1 ip 172.168.7.1
standby 1 priority 105
standby 1 preempt
standby 1 track Serial0/0/0
ipv6 enable
ipv6 add fe80::1 link-local
ipv6 add 172:DB8:ACAD:70::2/64
ipv6 nd other-config-flag
ipv6 dhcp server AD
exit

int fa0/0.99
encapsulation dot1Q 99
ip address 172.168.8.2 255.255.255.0
ip helper-address 172.168.10.2 255.255.255.0
standby 1 ip 172.168.8.1
standby 1 priority 105
```

```
standby 1 preempt
standby 1 track Serial0/0/0
ipv6 enable
ipv6 add fe80::1 link-local
ipv6 add 172:DB8:ACAD:99::2/64
ipv6 nd other-config-flag
ipv6 dhcp server MGT
exit

int fa0/0.150
encapsulation dot1Q 150
ip address 172.168.9.2 255.255.255.0
standby 1 ip 172.168.9.1
standby 1 priority 105
standby 1 preempt
standby 1 track Serial0/0/0
Exit
```

MAIN-MLS1-2

```
vlan 10
name LD
vlan 20
name MD
vlan 30
name HRD
vlan 40
name ITD
vlan 50
name VPD
vlan 60
name GDD
vlan 70
name AD
vlan 99
name MGT
```

```
vlan 150
name PHONE

interface range GigabitEthernet0/1-2
shutdown

interface range FastEthernet0/1-3
shutdown

interface range FastEthernet0/5-6
shutdown

interface range FastEthernet0/8-19
shutdown

interface FastEthernet0/21
shutdown

interface FastEthernet0/4
switchport trunk allowed vlan 1-1000
switchport trunk encapsulation dot1q
switchport mode trunk

interface FastEthernet0/7
switchport trunk allowed vlan 1-1000
switchport trunk encapsulation dot1q
switchport mode trunk

interface FastEthernet0/20
switchport trunk allowed vlan 1-1000
switchport trunk encapsulation dot1q
switchport mode trunk

interface FastEthernet0/22
switchport trunk allowed vlan 1-1000
```

```
switchport trunk encapsulation dot1q
switchport mode trunk

interface FastEthernet0/23
switchport trunk allowed vlan 1-1000
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode active

interface FastEthernet0/24
switchport trunk allowed vlan 1-1000
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode active

interface Port-channel2
switchport trunk allowed vlan 1-1000
switchport trunk encapsulation dot1q
switchport mode trunk
```

MAIN-SW

```
vlan 10
name LD
vlan 20
name MD
vlan 30
name HRD
vlan 40
name ITD
vlan 50
name VPD
vlan 60
name GDD
vlan 70
name AD
```

```
vlan 99
name MGT
vlan 150
name PHONE

interface Vlan99
ip address 172.168.8.4 255.255.255.0
no shutdown

interface range FastEthernet0/11-22
shutdown

interface GigabitEthernet0/1-2
shutdown

interface range FastEthernet0/1-5
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict

interface range FastEthernet0/6-10
switchport access vlan 99
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict

interface range FastEthernet0/6-7
switchport voice vlan 150

interface range FastEthernet0/1-2
switchport voice vlan 150
```

```
interface range FastEthernet0/23-24
switchport trunk allowed vlan 1-1000
switchport mode trunk
channel-group 1 mode active

interface Port-channel1
switchport trunk allowed vlan 1-1000
switchport mode trunk
```

Branch 1

B1RTR

```
hostname B1RTR
!
enable secret  cisco
banner motd "Warning Access to this device is strictly prohibited!"
service password-encryption
username admin password cisco
ip ssh version 2
ip domain-name cisco.com
crypto key generate rsa general-key mod 2048
aaa new-model
aaa authentication login default group tacacs+ local
tacacs-server host 192.1.1.2
tacacs-server key secretk3y
!
ip dhcp excluded-address 192.1.1.1 192.1.1.9
ip dhcp excluded-address 192.1.2.1 192.1.2.9
ip dhcp excluded-address 192.1.3.1 192.1.3.9
!
ip dhcp pool voice
 network 192.1.4.0 255.255.255.0
 default-router 192.1.4.1
```

```
 option 150 ip 192.1.4.1
ip dhcp pool vlan10
 network 192.1.1.0 255.255.255.0
 default-router 192.1.1.1
 dns-server 192.1.1.2
ip dhcp pool vlan20
 network 192.1.2.0 255.255.255.0
 default-router 192.1.2.1
 dns-server 192.1.1.2
ip dhcp pool vlan30
 network 192.1.3.0 255.255.255.0
 default-router 192.1.3.1
 dns-server 192.1.1.2
!
!
ipv6 unicast-routing
!
ipv6 dhcp pool Vlan10
 dns-server 2002:DB9:ACAD:1::2
!
ipv6 dhcp pool Vlan20
 dns-server 2002:DB9:ACAD:1::2
!
ipv6 dhcp pool Vlan30
 dns-server 2002:DB9:ACAD:1::2
!
interface FastEthernet0/1.10
 encapsulation dot1Q 10 native
 ip address 192.1.1.1 255.255.255.0
 ipv6 address FE80::1 link-local
 ipv6 address 2002:DB9:ACAD:1::1/64
 ipv6 nd other-config-flag
 ipv6 eigrp 10
 ipv6 address autoconfig
 ipv6 enable
```

```
 ipv6 dhcp server Vlan10
!
interface FastEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.1.2.1 255.255.255.0
 ipv6 address FE80::1 link-local
 ipv6 address 2002:DB9:ACAD:2::1/64
 ipv6 nd other-config-flag
 ipv6 eigrp 10
 ipv6 address autoconfig
 ipv6 enable
 ipv6 dhcp server Vlan20
!
interface FastEthernet0/1.30
 encapsulation dot1Q 30
 ip address 192.1.3.1 255.255.255.0
 ipv6 address FE80::1 link-local
 ipv6 address 2002:DB9:ACAD:3::1/64
 ipv6 nd other-config-flag
 ipv6 eigrp 10
 ipv6 address autoconfig
 ipv6 enable
 ipv6 dhcp server Vlan30
!
interface FastEthernet0/1.150
 encapsulation dot1Q 150
 ip address 192.1.4.1 255.255.255.0
 ipv6 address FE80::1 link-local
 ipv6 address 2002:DB9:ACAD:4::1/64
 ipv6 eigrp 10

interface Serial0/2/0
 ip address 60.60.60.1 255.255.255.252
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:FF::1/64
```

```
 ipv6 eigrp 10
 ipv6 enable

router eigrp 10
 eigrp router-id 1.0.0.1
 network 192.1.1.0
 network 192.1.2.0
 network 192.1.3.0
 network 192.1.4.0
 network 60.60.60.0 0.0.0.3
!
ipv6 router eigrp 10
 eigrp router-id 1.0.0.1
 no shutdown
!
telephony-service
 max-ephones 3
 max-dn 3
 ip source-address 192.1.4.1 port 2000
 auto assign 1 to 3
!
ephone-dn 1
 number 1101
!
ephone-dn 2
 number 1201
!
ephone-dn 3
 number 1301
!
line con 0
 password cisco
 login authentication default
!
line vty 0 4
```

```
 login authentication default
 transport input ssh
```

B1SW1

```
hostname B1SW1
enable secret  cisco
service password-encryption
ip default-gateway 192.1.1.1
banner motd "Warning Access to this device is strictly prohibited!"
ip ssh version 2
ip domain-name cisco.com
crypto key generate rsa general-keys mod 2048
tacacs-server host 192.1.1.2
tacacs-server key secretk3y

Vtp mod server
Vtp domain cisco.com
vlan10
vlan20
vlan30
vlan150

line con 0
 password cisco
 login authentication default

line vty 0 4
 password cisco
 login authentication default
 transport input ssh

aaa new-model
```

```
aaa authentication login default group tacacs+ local

username admin pass cisco

interface Port-channel1
 switchport trunk native vlan 10
 switchport trunk allowed vlan 1-1000
 switchport mode trunk

int ra fa0/1-8
sw mod acc
sw acc vlan 10

int ra fa0/9-10
sw mod tru
sw tru all vlan 1-1000
sw tru native vlan 10

int ra fa0/1-2
sw port-security
sw port-sec max 5
sw port-sec mac sticky
sw port-sec violation restrict
sw voice vlan 150

int ra fa0/11-12
sw port-security
sw port-sec max 5
sw port-sec mac sticky
sw port-sec violation restrict
sw voice vlan 150


int ra fa0/11-20
sw mod acc
```

```
sw acc vlan 20

int fa0/13
sw mod tru
no sw acc vlan 20
sw tru native vlan 10
sw tru all vlan 1-1000

int fa0/22-24
channel-group 1 mod active

int fa0/21
sw mod tru
sw tru all vlan 1-1000
sw tru native vlan 10

int po1
sw mod tru
sw tru all vlan 1-1000
sw tru native vlan 10

interface Vlan10
 ip address 192.1.1.6 255.255.255.0
```

Branch 2

B2RTR

```
hostname B2Rtr
enable secret cisco
service password-encryption
username admin password cisco
ip domain-name branch2rtr.com
crypto key generate rsa general-key mod 2048
banner motd "Authorized access only!!"
```

```
tacacs-server host 192.2.1.2
tacacs-server key secretpa55
line con 0
 login authentication default
line vty 0 4
 login authentication default
 transport input ssh
!
ip dhcp excluded-address 192.2.2.1 192.2.2.9
ip dhcp excluded-address 192.2.3.1 192.2.3.9
ip dhcp excluded-address 192.2.4.1 192.2.4.9
!
ip dhcp pool voice
 network 192.2.5.0 255.255.255.0
 default-router 192.2.5.1
 option 150 ip 192.2.5.1
ip dhcp pool vlan10
 network 192.2.2.0 255.255.255.0
 default-router 192.2.2.1
 dns-server 192.2.1.2
ip dhcp pool vlan20
 network 192.2.3.0 255.255.255.0
 default-router 192.2.3.1
 dns-server 192.2.1.2
ip dhcp pool vlan30
 network 192.2.4.0 255.255.255.0
 default-router 192.2.4.1
 dns-server 192.2.1.2
!
aaa new-model
!
aaa authentication login default group tacacs+ local
!
ipv6 unicast-routing
!
```

```
ipv6 dhcp pool Vlan10
 dns-server 2001:DB8:ACAD:1::2
!
ipv6 dhcp pool Vlan20
 dns-server 2001:DB8:ACAD:1::2
!
ipv6 dhcp pool Vlan30
 dns-server 2001:DB8:ACAD:1::2
!
interface FastEthernet0/1
 ip address 192.2.1.1 255.255.255.0
 duplex auto
 speed auto
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:1::1/64
 ipv6 nd other-config-flag
 ipv6 eigrp 10
 ipv6 address autoconfig
 ipv6 enable
!
interface FastEthernet0/1.10
 encapsulation dot1Q 10
 ip address 192.2.2.1 255.255.255.0
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:2::1/64
 ipv6 nd other-config-flag
 ipv6 eigrp 10
 ipv6 address autoconfig
 ipv6 enable
 ipv6 dhcp server Vlan10
!
interface FastEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.2.3.1 255.255.255.0
 ipv6 address FE80::1 link-local
```

```
 ipv6 address 2001:DB8:ACAD:3::1/64
 ipv6 nd other-config-flag
 ipv6 eigrp 10
 ipv6 address autoconfig
 ipv6 enable
 ipv6 dhcp server Vlan20
!
interface FastEthernet0/1.30
 encapsulation dot1Q 30
 ip address 192.2.4.1 255.255.255.0
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:4::1/64
 ipv6 nd other-config-flag
 ipv6 eigrp 10
 ipv6 address autoconfig
 ipv6 enable
 ipv6 dhcp server Vlan30
!
interface FastEthernet0/1.150
 encapsulation dot1Q 150
 ip address 192.2.5.1 255.255.255.0
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:5::1/64
 ipv6 nd other-config-flag
 ipv6 eigrp 10
 ipv6 address autoconfig
 ipv6 enable
!
interface Serial0/2/0
 ip address 50.50.50.1 255.255.255.252
 ipv6 address FE80::1 link-local
 ipv6 address 2001:DB8:ACAD:F::1/64
 ipv6 eigrp 10
 ipv6 enable
 clock rate 2000000
```

```
!
router eigrp 10
 network 192.2.1.0
 network 192.2.2.0
 network 192.2.3.0
 network 192.2.4.0
 network 192.2.5.0
 network 50.50.50.0 0.0.0.3
!
ipv6 router eigrp 10
 eigrp router-id 2.1.1.1
 no shutdown
!
!
telephony-service
 max-ephones 3
 max-dn 3
 ip source-address 192.2.5.1 port 2000
 auto assign 1 to 3
!
ephone-dn 1
 number 2101
!
ephone-dn 2
 number 2201
!
ephone-dn 3
 number 2301
!
end
```

B2SW1

```
hostname B2S1
enable secret cisco
service password-encryption
```

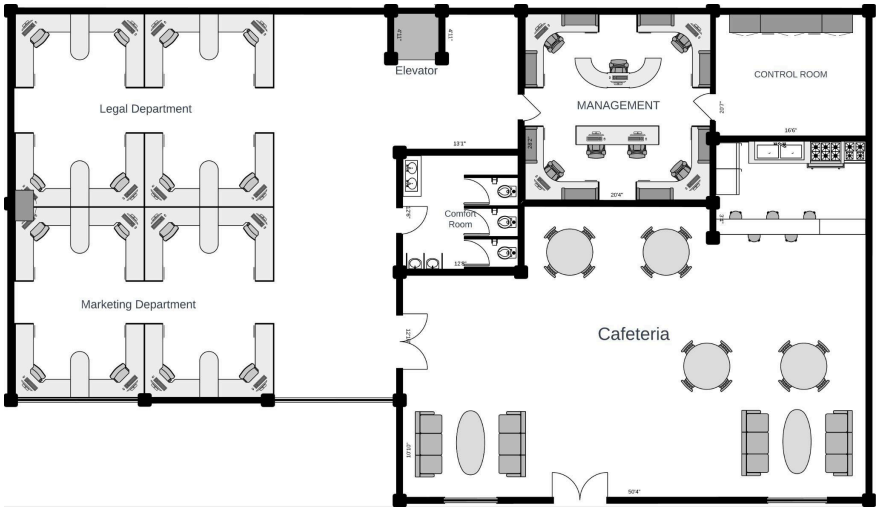
```
banner motd "Authorized access only !!"
tacacs-server host 192.2.1.2
tacacs-server key secretpa55
aaa new-model
aaa authentication login default group tacacs+ local
ip ssh version 2
ip domain-name B2sw1.com
crypto key generate rsa general-key mod 2048
username admin password cisco
!
vlan 10
vlan 20
vlan 30
vlan 150
!

int ra fa0/1-2
sw mod acc
sw acc vlan 10
sw port-sec
sw port-sec max 5
sw port-sec mac sticky
sw port-sec violation restrict
sw voice vlan 150
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
 shutdown
!
int ra fa0/22-24
channel-group 1 mode active
!
interface Port-channel1
 switchport trunk allowed vlan 1-1000
```

```
 switchport mode trunk
!
interface GigabitEthernet0/1
 switchport trunk allowed vlan 1-1000
 switchport mode trunk
!
interface Vlan1
 ip address 192.2.1.3 255.255.255.0
!
ip default-gateway 192.2.1.1
!
line con 0
 password cisco
 login authentication default
!
line vty 0 4
 login authentication default
 transport input ssh
!
end
```

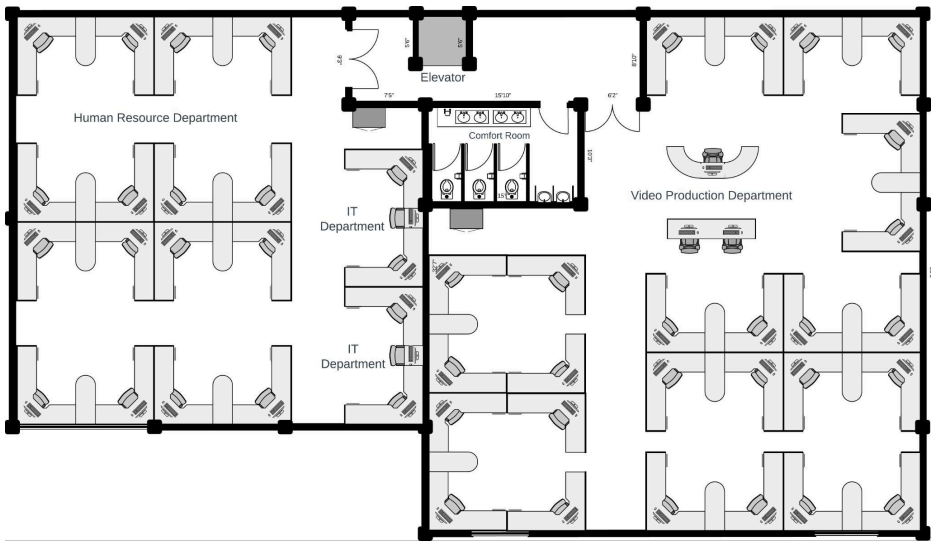
Floorplan

Main Branch

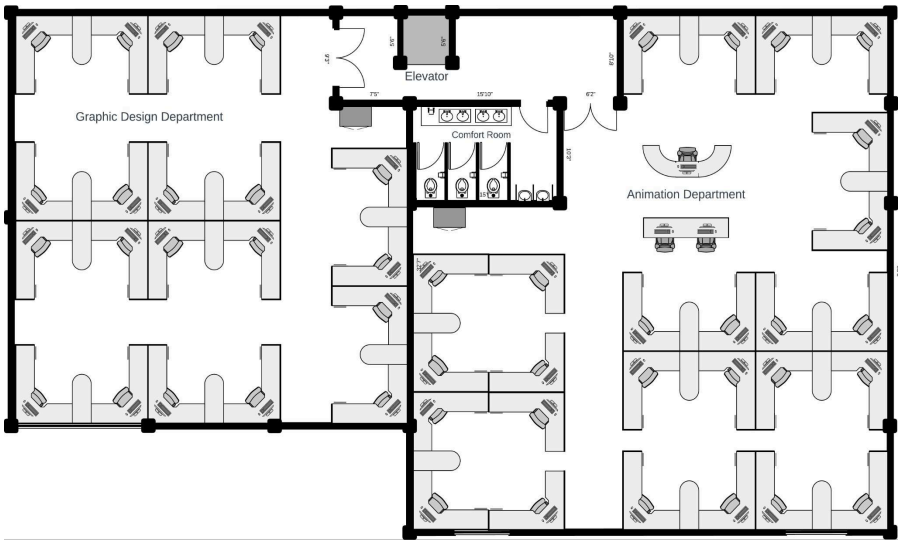
1st Floor



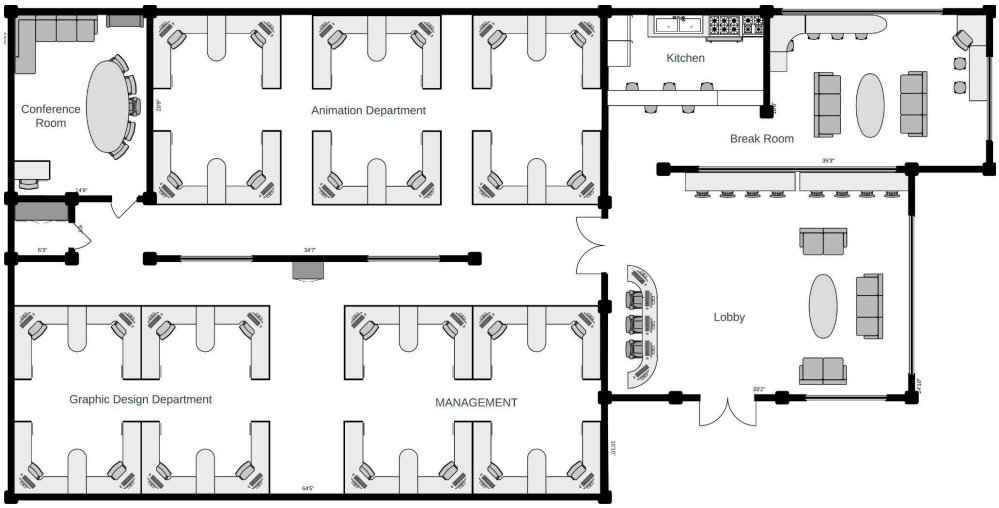
2nd Floor



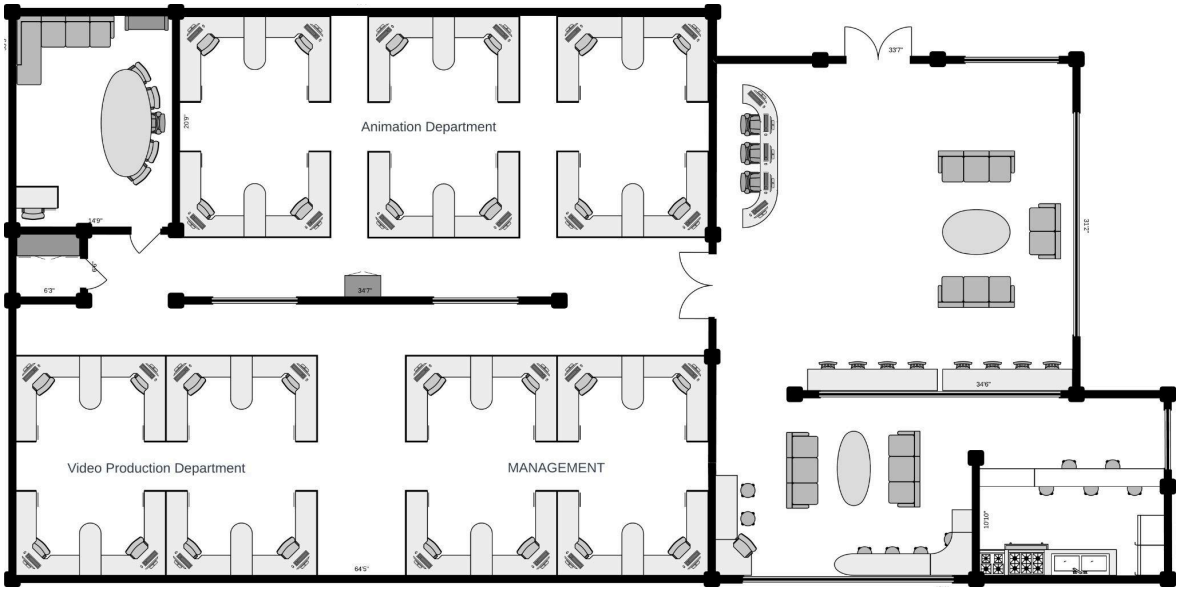
3rd Floor



Branch 1



Branch 2


Animation Department
Video Production Department
MANAGEMENT
33'7"
31'2"
34'6"
34'7"
14'9"
6'3"
64'5"
10'10"

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